Trotter layers by addition into a phase gradient

Samuel J. Elman[1]

[1] *University of Technology Sydney, Sydney, Australia*

(Dated: November 9, 2026)

A Trotter step of a Hamiltonian with many distinct coefficients is a product of layers of commuting Pauli rotations, and a fault-tolerant compilation synthesises each rotation on its own. The best Clifford+T synthesis is a probabilistic mixture: about $0.53\log_2(1/\epsilon)$ T gates for a channel within diamond error ϵ, whatever the angle, and, as Bothe *et al.* have recently shown, far fewer when $\theta^2 \lesssim \epsilon$. We ask whether a catalytic phase-gradient register can do better. For one rotation it cannot: adding the angle into a b-qubit gradient costs $b-2$ Toffoli gates, about $4\log_2(1/\epsilon)$ T gates. For a layer it can. We write every angle of the layer in signed binary digits, reduce the digits column by column with full adders, and add the one bit left in each column into the gradient register once. A layer of L rotations with N signed digits then costs at most $4N + 4(b-2)$ T gates, plus one for each digit worth $\pi/4$, so a rotation by θ costs about $\frac{4}{3}\log_2(\theta/\Delta)$ T gates on a grid of spacing Δ: the cost falls with the angle across the whole range $\sqrt{\epsilon} \lesssim \theta \lesssim 1$, where synthesis does not. Rounding each angle at random between two grid points leaves only dephasing, so the grid can be coarse, $\Delta \approx \sqrt{2\epsilon}$, and rounding onto a sparse pair of grid points costs about $16\theta^2/\epsilon$ T gates in expectation when $\theta^2 < \epsilon/2$. We prove the counts and the errors and simulate every construction exactly, including the measurement-based uncomputation. For Heisenberg chains of 64 spins in a random field, run to time 10 at total errors of 10^{-3} and 10^{-1} with Trotter steps from 0.3 to 10^{-6}, the gradient layers use 0.16 to 0.88 times the T gates of the angle-dependent synthesis of Bothe *et al.*, and never more; nearest rounding, which gives a unitary, uses 0.24 to 0.51 times the T gates of gridsynth. The price is ancillas: at most 589 for the randomised layers.

I. INTRODUCTION

Product formulas are the simplest way to simulate a Hamiltonian $H = \sum_j h_j P_j$, with P_j Pauli strings, on a quantum computer [1–3]. One step of length δt is a product of rotations $e^{-i\theta_j P_j}$ with $\theta_j = h_j \delta t$ (or a fixed fraction of it), arranged in layers of commuting terms. In a fault-tolerant computation built from Clifford gates and T gates the rotations are the expensive part, and the standard treatment synthesises each one to its share of the error budget. Ross and Selinger's algorithm [4] gives a unitary with about $3\log_2(1/\epsilon)$ T gates. Kliuchnikov, Lauter, Minko, Paetznick and Petit [5] improved and combined the two ways of doing better, fallback circuits with an ancilla and a measurement [6] and probabilistic mixtures of approximations [7, 8]: their mixed fallback protocol uses on average $0.53\log_2(1/\epsilon) + 4.86$ T gates for a channel within diamond error ϵ, and their mixed diagonal protocol, with no ancilla, $1.52\log_2(1/\epsilon) - 0.01$. These counts do not depend on the angle. That is odd for Trotter: as the step shrinks every angle shrinks with it, and a rotation by 10^{-5} to precision 10^{-10} is charged the same as a rotation by one.

Bothe, Sünderhauf, Witham, Campbell and Blunt [9] have recently removed this oddity in part. In a mixture the cheap end can be the identity, so a small rotation is mostly not applied at all, and when $\theta^2 \lesssim \epsilon$ the expected cost falls to $\tilde{O}(\theta^2/\epsilon)$; the T count of a Trotter circuit can then stay constant as the step goes to zero. When $\theta^2 \gg \epsilon$ their cost returns to that of Ref. [5]. Their cost formula, built on the protocols of Ref. [5], is the comparison in this paper.

The question of this paper is whether a phase-gradient register helps. The register $|\Psi_b\rangle = 2^{-b/2}\sum_y e^{-2\pi i y/2^b}|y\rangle$ is an eigenstate of every addition modulo 2^b, so adding an integer K into it applies the phase $e^{2\pi i K/2^b}$ and leaves the register as it was [10, 11]. Additions cost Toffoli gates, not rotations, and the register is prepared once for the whole computation. Phase gradients are used this way for the quantum Fourier transform [12], for rotations by computed angles [13], and for rotations by equal angles after Hamming-weight phasing [11, 14]. For a layer of rotations by distinct, fixed angles, the usual view is that each needs its own controlled addition, about b Toffoli gates, and so loses to synthesis.

Our answer has two halves.

For one rotation, no. Adding θ into a b-qubit gradient costs $b - 2 - p_0$ Toffoli gates, p_0 the position of the lowest digit of the angle (Proposition 2), about $4\log_2(1/\epsilon)$ T gates against gridsynth's $3.03\log_2(1/\epsilon) + 5.4$, and, with randomised rounding, about $2\log_2(1/\epsilon)$ against $0.53\log_2(1/\epsilon) + 4.86$.

For a layer, yes. The phase of a layer, $\sum_j \theta_j x_j$ on bits x_j, is a weighted sum of bits, and once the angles are written in signed binary digits it is a sum of columns of bits of equal weight. A column can be reduced with full adders, one Toffoli for every bit removed, exactly as Hamming-weight phasing reduces one column of equal angles. After the reduction one bit is left per column, and those bits are added into the gradient register once, for $b-2$ Toffoli gates shared by the whole layer (Theorem 1). A rotation then costs about four T gates per signed digit of its angle. Its angle has about $\frac{1}{3}\log_2(\theta/\Delta)$ nonzero digits on a grid of spacing Δ, so the cost falls with the angle; and since rounding at random to the grid leaves only dephasing (Lemma 4), Δ can be as coarse as $\sqrt{2\epsilon}$.

So the gradient's cost falls with the angle over the whole range $\sqrt{\epsilon} \lesssim \theta \lesssim 1$, the range in which the cost of Ref. [9] does not. For $\theta \lesssim \sqrt{\epsilon}$, rounding at random onto a sparse pair of grid points gives the same θ^2/ϵ scaling as Ref. [9] (Proposition 5). With batches sized by their digits rather than by their rotations, the gradient stays the cheaper down to about $\sqrt{\epsilon}/10$; below that, synthesis is, and both cost a few hundredths of a T gate per rotation.

Section II sets up the layer and the gradient. Section III gives the circuit and its cost, Sec. IV the rounding and its error, and Sec. V the counts, against the angle-dependent formulas of Ref. [9] evaluated on the same angles, and against gridsynth, for single layers and for Heisenberg chains run to a fixed time. Section VI says what the comparison leaves out. The code, which compiles each layer to Clifford gates, T gates and AND gates and simulates it exactly, is available [15].

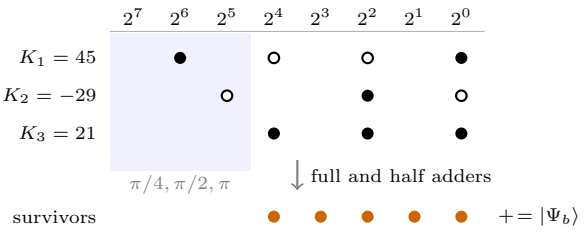


FIG. 1. The phase of a layer as columns of bits ($b = 8$). Each rotation contributes one bit per nonzero signed digit of its angle: a filled dot is x_j, an open dot is $\bar{x}_j$ (a digit -1, $-x = \bar{x} - 1$). Columns $b-3$, $b-2$, $b-1$ (shaded) are worth $\pi/4$, $\pi/2$ and π per bit and get T, S and Z gates directly. The other columns are reduced from the right to one bit each, carries moving left, with one AND per full or half adder, and the survivors are added into the gradient register as one integer.

II. SETTING

A layer. A set of commuting Pauli strings can be mapped by a Clifford circuit to strings of Zs, and the parity of a string of Zs can be computed into one qubit with CNOTs. So a layer is, up to Clifford gates,

$$V(\boldsymbol{\theta}) = \prod_{j=1}^{L} e^{-i\theta_j Z_j} = e^{-i\sum_j \theta_j} \operatorname{diag}_x e^{2i\sum_j \theta_j x_j}, \qquad (1)$$

with Z_j acting on a bit x_j (a qubit, or a computed parity) and $Z_j = 1 - 2x_j$. Several Z_j may share a bit; nothing below needs them to be distinct. We drop the global phase throughout. When the layer is controlled, as in phase estimation, the global phase becomes a phase on the control and costs one more addition.

The gradient. For $b \geq 4$ let $|\Psi_b\rangle = 2^{-b/2} \sum_{y=0}^{2^b-1} e^{-2\pi i y/2^b} |y\rangle$. If a circuit A maps $|x\rangle|y\rangle$ to $|x\rangle|y + f(x) \bmod 2^b\rangle$, then

$$A|x\rangle|\Psi_b\rangle = e^{2\pi i f(x)/2^b} |x\rangle|\Psi_b\rangle . \qquad (2)$$

The register is a product state: qubit k is $(|0\rangle + e^{-2\pi i 2^k/2^b}|1\rangle)/\sqrt{2}$, so the top qubit is $|-\rangle$, the next $S^\dagger|+\rangle$, the next $T^\dagger|+\rangle$. It is prepared once, with $b-3$ synthesised rotations, and returned unchanged by every addition. Its preparation error enters once, not once per use, because the whole computation is a fixed channel applied to the input and the register. On the gradient, the rotation $e^{-i\theta Z}$ on a bit x is the addition of Kx with

$$\theta = K\Delta, \qquad \Delta = \pi/2^b, \qquad (3)$$

so the Z angles live on a grid of spacing Δ.

Gates and costs. We count T gates. The circuits use Clifford gates, T gates, and the logical AND of Jones [16] and Gidney [11]: a Toffoli onto a fresh qubit, four T gates, uncomputed by an X-basis measurement and a CZ conditioned on the outcome, with no T gate. Every AND below is uncomputed this way, so a Toffoli costs four T gates once. In a surface-code factory a Toffoli is cheaper than four T gates [17]; counting four is the conservative choice for the gradient.

Errors. Errors are diamond distances [18], per rotation, as in the synthesis literature. For two unitaries that differ by $e^{-i\delta Z}$ the distance is $2|\sin\delta|$. Unitary errors add along a circuit, and so do channel errors in the diamond norm.

Signed digits. Every integer has a unique non-adjacent form (NAF) $K = \sum_p d_p 2^p$ with $d_p \in \{-1, 0, 1\}$ and no two adjacent digits nonzero [19]. It has the fewest nonzero digits of any signed-digit expansion of K [19, 20]; we write $\mathrm{wt}(K)$ for their number. An integer below 2^ℓ in magnitude has $\mathrm{wt}(K) \leq \lceil(\ell+1)/2\rceil$, and a random one about $\ell/3$ [20]. Modulo 2^b we expand the representative of K in $(-2^{b-1}, 2^{b-1}]$; a digit at position b would be worth a whole turn and is dropped.

III. ONE ADDITION PER LAYER

Write each K_j in signed digits and each digit $d_{jp} = -1$ on a bit x as the bit $\bar{x}$ with the constant -1. Then

$$\sum_j K_j x_j + C \equiv \sum_{p=0}^{b-1} 2^p \left|\{\text{ones in column } p\}\right| \pmod{2^b}, \qquad (4)$$

where column p holds one bit, x_j or $\bar{x}_j$, for each nonzero digit d_{jp}, and C is the sum of 2^p over the digits -1 (Fig. 1). The constant only changes the global phase. The circuit has three parts.

Compute. Copy the bit of each digit at a position $p \leq b-4$ into a fresh qubit in column p (a CNOT, and an X for a digit -1). Then, for $p = 0, 1, \ldots, b-4$ in turn, reduce column p to at most one bit. While it holds three or more bits, replace three of them, a, a' and c, by their sum $a \oplus a' \oplus c$ in column p and their majority in column

$p+1$:

$$a + a' + c = (a \oplus a' \oplus c) + 2\,\mathrm{maj}(a, a', c), \tag{5}$$

computed in place with CNOTs and one AND, since $\mathrm{maj}(a, a', c) = a \oplus (a \oplus a')(a \oplus c)$. When two bits are left, replace them by $a \oplus a'$ in column p and aa' in column $p+1$, again one AND.

Phase. Apply T, S and Z to every bit in columns $b-3$, $b-2$ and $b-1$: the bits of the digits there, and the carries out of column $b-4$. Add the survivors, one bit in each column $p \le b-4$, as an integer $S < 2^{b-3}$ into the gradient register, with Gidney's adder [11]. Because the top qubit of the register is $|-\rangle$, the carry into it is applied as a phase and never computed.

Uncompute. Run the compute part backwards, each AND undone by measurement.

Theorem 1 (a layer)**.** *Let $b \ge 4$ and let $K_1, \ldots, K_L$ be integers whose signed-digit expansions modulo 2^b have N nonzero digits at positions $p \le b-4$ and N_3 at position $b-3$. The circuit above, with one copy of $|\Psi_b\rangle$, implements $|x\rangle \mapsto e^{2\pi i(\sum_j K_j x_j + C)/2^b}|x\rangle$ exactly, returns the register and every ancilla unchanged, and uses at most $N + b - 2$ AND gates and $N_3 + c$ further T gates, where $c \le N/2$ is the number of carries out of column $b-4$. Its T count is at most*

$$\mathcal{T} \le 4N + N_3 + 4(b-2). \tag{6}$$

It uses at most $2N + b - 2$ ancillas besides the register, plus one for each digit -1 in the top three columns.

Proof. Correctness. Each full or half adder keeps the right-hand side of Eq. (4) fixed, since it replaces bits of total value $a + a' + c$ (or $a + a'$) in column p by bits of the same value in columns p and $p+1$. The T, S and Z gates on columns $b-3$, $b-2$ and $b-1$ apply $e^{2\pi i 2^p/2^b}$ for each bit set there, and the addition applies $e^{2\pi i S/2^b}$ by Eq. (2). Together they apply $e^{2\pi i(\sum_j K_j x_j + C)/2^b}$. The compute part is a reversible classical map with all inputs kept (each AND target and each copy is fresh), so running it backwards restores every ancilla to $|0\rangle$; the measured uncompute of an AND is exact [11].

Counts. Let column p receive n_p digit bits and c_p carries, $c_0 = 0$, and $m_p = n_p + c_p$. A full adder removes two bits from the column and a half adder one, and each sends one carry on, so reducing m_p bits to one takes $\lfloor m_p/2 \rfloor$ adders and $c_{p+1} = \lfloor m_p/2 \rfloor$. The ANDs of the compute part number $A = \sum_{p=0}^{b-4} c_{p+1}$, and

$$A \le \sum_{p=0}^{b-4} \frac{n_p + c_p}{2} = \frac{N + A - c_{b-3}}{2}, \tag{7}$$

so $A \le N - c_{b-3}$. The adder for S has one AND per carry position from the lowest survivor p_0 to $b-3$, that is $b-2-p_0 \le b-2$, and its top carry is a phase. The direct gates add $N_3 + c_{b-3}$ T gates (S and Z are Clifford). So $\mathcal{T} \le 4(N - c_{b-3}) + 4(b-2) + N_3 + c_{b-3} \le 4N + N_3 + 4(b-2)$. *Ancillas:* N copies, $A \le N$ carries, at most $b-2$ in the adder, and a copy for each negated top digit. □

Hamming-weight phasing [11, 14] is the case of one column: L bits that share an angle are added into a $\lfloor \log_2 L \rfloor + 1$-bit count with about L ANDs, and the count is rotated. The construction above is the same reduction applied to every column of the angles' digits at once, the reduction of a multiplier's partial products [21], with the gradient register as the final adder. Equal angles can be combined with it: our code first sums a class of bits that share an angle into its Hamming weight $W = \sum_k w_k 2^k$, and then treats each bit w_k as one rotation, by $2^k\theta$ or by its own rounding of it.

Proposition 2 (one rotation)**.** *For $L = 1$ and K with lowest nonzero digit at position $p_0 \le b-4$, the circuit has no adders in the compute part and $b-2-p_0$ ANDs in the addition, so $\mathcal{T} = 4(b-2-p_0) + N_3$.*

Proof. Each column holds at most one bit, so the compute part only copies. The rest is the count of the adder in the proof of Theorem 1. □

The proposition is also the cost of the controlled constant addition that rotates one qubit: fanning the bit out to the positions of the digits of K with CNOTs makes it an ordinary addition, so the control costs nothing extra. For a generic angle p_0 is small, and at the error ϵ of a unitary rounding (Sec. IV), $b = \lceil \log_2(\pi/\epsilon) \rceil$ and $\mathcal{T} \approx 4\log_2(1/\epsilon)$, against $3.03\log_2(1/\epsilon) + 5.4$ for gridsynth. For channels the comparison is worse: $\mathcal{T} \approx 2\log_2(1/\epsilon)$ against $0.53\log_2(1/\epsilon) + 4.86$. A phase gradient on its own does not make an isolated rotation cheaper. What it does is make the $b-2$ Toffoli gates of the carry chain a cost per layer rather than per rotation.

Corollary 3 (cost per rotation)**.** *Split a layer into batches of B rotations, each with its own compute part and addition. A rotation costs at most $4\,\mathrm{wt}(K) + 4(b-2)/B + [\text{digit at } b-3]$ T gates, its share of the addition included. If $|\theta| \le 2^\ell \Delta$, then $\mathrm{wt}(K) \le \lceil (\ell+2)/2 \rceil$.*

So the cost per rotation is set by the number of nonzero digits of its angle, which grows as $\log_2(\theta/\Delta)$ and not as $\log_2(1/\Delta)$: the digits below position $\log_2 \Delta$ are not there, and those above $\log_2 \theta$ are zero.

IV. ROUNDING

The integers K_j approximate θ_j/Δ. There are two ways to choose them.

Nearest. $K = \mathrm{round}(\theta/\Delta)$ leaves a rotation error $|\delta| \le \Delta/2$, a diamond error $2\sin(\Delta/2) \le \Delta$. A per-rotation error ϵ needs the least b with $2\sin(\pi/2^{b+1}) \le \epsilon$, $b \approx \log_2(\pi/\epsilon)$.

Randomised. Choose between the two neighbours at random, with the probability that cancels the coherent part of the error.

Lemma 4 (randomised rounding)**.** *Let $u, v \geq 0$, $0 < u+v \leq \pi/2$, and let $q = \sin 2u/(\sin 2u + \sin 2v)$. The channel that applies $e^{-i(\theta+v)Z}$ with probability q and $e^{-i(\theta-u)Z}$ with probability $1-q$ is $\mathcal{U}_\theta \circ \mathcal{D}_p$, where $\mathcal{U}_\theta$ is the rotation by θ, $\mathcal{D}_p(\rho) = (1-p)\rho + pZ\rho Z$, and*

$$p = \frac{\sin u \sin v}{\cos(u-v)} = \frac{1}{2}\Big(1 - \frac{\cos(u+v)}{\cos(u-v)}\Big) \leq \sin^2 \frac{u+v}{2}. \quad (8)$$

Its diamond distance from $\mathcal{U}_\theta$ is $2p$.

Proof. For a rotation by δ after $\mathcal{U}_\theta$, $e^{-i\delta Z}\rho e^{i\delta Z} = \cos^2\delta\,\rho + \sin^2\delta\, Z\rho Z - \frac{i}{2}\sin 2\delta\,[Z,\rho]$. Averaged over $\delta = v$ with probability q and $\delta = -u$ with probability $1-q$, the commutator term has weight $q\sin 2v - (1-q)\sin 2u = 0$, and the rest is $\mathcal{D}_p$ with $p = (1-q)\sin^2 u + q\sin^2 v$. Substituting q and using $\sin 2x = 2\sin x\cos x$ gives $p = \sin u \sin v\, \sin(u+v)/[\sin(u+v)\cos(u-v)]$, and $2\sin u\sin v = \cos(u-v) - \cos(u+v)$ gives the second form; it is largest at $u = v$. Finally $\|\mathcal{D}_p - \mathrm{id}\|_\diamond = 2p\,\big\|\frac{1}{2}(Z\cdot Z - \mathrm{id})\big\|_\diamond = 2p$, the norm attained on a maximally entangled input. □

With $u+v = \Delta$ the dephasing is at most $\sin^2(\Delta/2)$ and the error at most $2\sin^2(\Delta/2) \approx \Delta^2/2$: the grid can be coarser, $b \approx \frac{1}{2}\log_2(\pi^2/2\epsilon)$, half the bits of nearest rounding. This is the mixing of Hastings [7] and Campbell [8] applied to the grid, where it is exact: the two neighbours differ only in their Z angle, so no off-axis error survives the average. In a layer the roundings are drawn independently, the circuit compiled for each draw implements its rotations exactly (Theorem 1), and the averaged channel is $V(\boldsymbol{\theta})$ followed by independent dephasings by the operators Z_j. These commute, so the layer's error is at most $\sum_j 2p_j$. The same holds for a rotation of a computed bit, such as a bit w_k of a Hamming weight: $(-1)^{w_k(x)}$ is a diagonal involution and the proof is unchanged. The draws are classical and made at compile time; each Trotter step draws afresh.

Sparse pairs. Nothing in Lemma 4 needs u and v to be one grid step. Rounding onto a farther pair $K_- < \theta/\Delta < K_+$ costs more dephasing and may cost fewer digits. Our code searches, for each weight w, the nearest grid points of weight at most w on either side and the farthest partner each can have within the dephasing budget, and takes the pair with the least expected weight $(1-q)\,\mathrm{wt}(K_-) + q\,\mathrm{wt}(K_+)$; the adjacent pair is among the candidates. For moderate angles this saves little (Sec. V). For small angles it is what matters.

Proposition 5 (small angles)**.** *Let $\theta > 0$, $\theta^2 \leq \epsilon/2$ and $\sin^2(\Delta/2) \leq \epsilon/2$. Let $D = 2^k\Delta$ with k the largest integer for which $D \geq \theta$ and the pair $\{0, D\}$ has dephasing at most $\epsilon/2$, and suppose $2D \leq \pi/4$. Randomised rounding onto $\{0, D\}$ has error at most ϵ, and uses one digit with probability*

$$q \leq \frac{4\theta^2}{\epsilon} \cdot \frac{1}{\cos 2D\,(1 - 2D^2/3)} \quad (9)$$

and none otherwise. Its expected T count is at most $4q$, that is $16\theta^2/\epsilon$ to leading order, plus its share of an addition that is needed only if some digit of the batch is drawn.

Proof. Such a k exists: let D_0 be the least $2^j\Delta \geq \theta$. If $D_0 = \Delta$ the dephasing of $\{0, D_0\}$ is at most $\sin^2(\Delta/2) \leq \epsilon/2$ by Lemma 4; otherwise $D_0 < 2\theta$ and it is at most $\sin^2(D_0/2) < \theta^2 \leq \epsilon/2$. By maximality the pair $\{0, 2D\}$ has dephasing above $\epsilon/2$: $\sin\theta\sin(2D-\theta) > \frac{\epsilon}{2}\cos(2D - 2\theta) \geq \frac{\epsilon}{2}\cos 2D$, and with $\sin x \leq x$, $D > \epsilon\cos 2D/(4\theta)$. Next, $q = \sin 2\theta/(\sin 2\theta + \sin 2(D-\theta))$, and since $\sin a + \sin b \geq \sin(a+b)$ for $a, b \geq 0$, $a+b \leq \pi$, $\sin 2\theta + \sin 2(D-\theta) \geq \sin 2D \geq 2D(1 - 2D^2/3)$. So $q \leq \theta/[D(1-2D^2/3)]$, and the bound on D gives the result. The error is $2p \leq \epsilon$ by Lemma 4. A drawn digit costs at most four T gates: by the proof of Theorem 1 it adds at most one AND to the compute part, or one T gate if it lands in column $b-3$. □

When $\theta < \epsilon/2$ the rotation can be dropped altogether, at error $2\sin\theta < \epsilon$. Proposition 5 is the gradient's version of the small-angle regime of Ref. [9]: in both, the cost of a rotation with $\theta^2 \ll \epsilon$ is of order θ^2/ϵ, because a rotation that small is mostly not applied. The gradient's expected digit cost, $16\theta^2/\epsilon$, is below theirs, which at $\theta = \sqrt{\epsilon}/10$ is between 30 and 50 times θ^2/ϵ; what decides the comparison there is the share of the addition, paid by a batch only if it draws a digit (Sec. V). Between the two regimes, for $\epsilon \ll \theta^2$, the gradient's cost is about $\frac{4}{3}\log_2(\theta/\Delta) \approx \frac{2}{3}\log_2(\theta^2/2\epsilon)$ T gates with randomised rounding, and still falls with the angle; theirs is $\min\{0.53\log_2(1/\epsilon) + 4.86,\ \dots\}$ and in this regime equal to the first term.

Sparse nearest rounding. The same search for nearest rounding (the lightest grid point within the error, on a grid finer than needed) changes the mean weight by less than one per cent for every angle scale we tried, and we do not use it.

V. COUNTS

The gradient. The gradient layers are compiled by our code to Clifford gates, T gates and ANDs, and every compiled circuit is checked by exact simulation: with the system in each basis state and the gradient register in $|\Psi_b\rangle$, the circuit maps basis states to phases times basis states, so all 2^b branches of the register can be followed at once and the output compared with $e^{2\pi i(f(x)+C)/2^b}\,|x\rangle\,|\Psi_b\rangle$, ancillas returned to $|0\rangle$. The tests do this for 100 random layers with up to eight bits and $b \leq 10$, including equal-angle classes and per-bit roundings, and a full state-vector simulation with the ANDs un-

computed by actual X measurements with random outcomes agrees to 10^{-16}. The bound (6) is checked on every layer counted below.

The layers are split into batches of at most $D = 256$ signed digits, not of a fixed number of rotations. The ancillas grow with the digits of a batch (Theorem 1), so this bounds them, to between 500 and 660 in every run below, whatever the angles; and rotations with few or no digits, the small ones, fill large batches, so the addition's $b-2$ Toffolis are shared widely where the digits are sparse.

Synthesis. The comparison is the angle-dependent cost of Bothe *et al.* [9], which we evaluate on the same angles as the gradient. We transcribe their costing script (their Appendix A): an exactly enumerated staircase of over-rotations up to 35 T gates, and beyond it their asymptotic formula, both with the identity as the under-rotation. Their two schemes are

$$\mathcal{T}_{\rm af}(\theta,\epsilon) = \min\{T_{\rm sa}(\theta,\epsilon),\ 1.52\log_2(1/\epsilon) - 0.01\}, \quad (10)$$
$$\mathcal{T}_{\rm fb}(\theta,\epsilon) = \min\{T_{\rm sa}(\theta,2\epsilon),\ 0.53\log_2(1/\epsilon) + 4.86\}, \quad (11)$$

ancilla-free and fallback, where $T_{\rm sa}$ is the small-angle cost and the second terms are the mixed diagonal and mixed fallback fits of Kliuchnikov *et al.* [5] (their Table 1, Clifford+T, mean T count). The fallback scheme is the cheaper of the two at every angle, and is the one we compare with; Eq. (11) is their Eq. (15). Their costs are for probability mixtures with diamond error ϵ, which is what the gradient's randomised rounding gives. They also show that the same counts hold for quasi-probability mixtures with $\lambda - 1 = \epsilon$, and their own Trotter estimates use those with a total $\lambda \approx e^2$, a sampling overhead rather than an error; we do not compare in that setting, since the gradient layers as built here are channels. The counts are their formulas, not runs of their method; their staircase is exact and their asymptotic formula derived, and they show it matches their data.

For a unitary with no mixing, the reference is gridsynth [4], run (pygridsynth 2.0 [22]) to operator-norm error $\epsilon/2$, which is diamond error ϵ, on 40 random angles at each $\epsilon = 10^{-3}, \ldots, 10^{-15}$: the mean T count is $3.03\log_2(1/\epsilon) + 5.4$, the $3.02\log_2(1/\epsilon) + 1.77$ of Ref. [5] up to the offset. Run on angles from 0.7 down to 7×10^{-7} at $\epsilon = 10^{-10}$ it uses 106 to 107 T gates whatever the angle. Nearest rounding is compared with it. The unmixed fallback protocol of Ref. [5], $1.03\log_2(1/\epsilon) + 5.75$, is cheaper than gridsynth and is the stronger reference for a coherent error; we quote it where it matters.

One layer. Figure 2 shows the cost per rotation in a layer of 1,024 rotations. At $\epsilon = 10^{-10}$, randomised rounding onto sparse pairs ($b = 18$) uses 20.5 T gates per rotation for angles of order one, 16.2 at 10^{-2}, 7.3 at 10^{-4} and 2.6 at 10^{-5}. Bothe *et al.*'s fallback scheme uses 22.5 at all of these but the last, where it is 18.9, and their ancilla-free scheme 50.5 and 30.6. Below $\theta \approx \sqrt{\epsilon} = 10^{-5}$ both costs collapse: 0.65 against 2.84 at 3×10^{-6}, 0.09 against 0.28 at 10^{-6}, and then 0.05 against 0.02 at 3×10^{-7}, where theirs is the smaller. At $\epsilon = 10^{-14}$ and 10^{-6} the picture is the same; the crossing, where there is one, lies below $\sqrt{\epsilon}/10$. So the gradient wins by a factor that grows as the angle falls from one towards $\sqrt{\epsilon}$, and synthesis wins well below it, at a few hundredths of a T gate. For angles of order one the two are close: 9 per cent apart at $\epsilon = 10^{-10}$, and at $\epsilon = 10^{-14}$ the gradient is 4 per cent worse (30.7 against 29.5). Plain randomised rounding, onto adjacent grid points, agrees with sparse pairs down to $\sqrt{\epsilon}$ and costs more below (0.31 against 0.09 at 10^{-6}).

Nearest rounding ($b = 35$) uses 45.6 T gates per rotation at angles of order one, 41.0 at 10^{-2}, 32.2 at 10^{-4} and 23.0 at 10^{-6}, against 106 for gridsynth and 40.0 for the unmixed fallback protocol. Its cost falls by about one T gate per halving of the angle, a little below the 4/3 of the digit count because the digits in the top three columns cost one T gate or none.

Batches and qubits. The addition's $b-2$ Toffolis are shared by a batch, and the ancillas grow with it (Fig. 3). With $B = 1$, nearest rounding costs 128 T gates against gridsynth's 106, and randomised rounding 59.7 against 22.5 for synthesis: Proposition 2 in numbers. With $B = 2$ nearest rounding is already at 73; with $B = 16$ the costs are 46.8 and 21.8 with 389 and 187 ancillas, and with $B = 64$, 43.2 and 20.2 with 1,402 and 662. Beyond $B = 64$ nothing is gained. Batches of 256 digits, as in the other runs, cost 45.6 and 20.5 with 541 and 527 ancillas. The ancillas are the copies of the digit bits and the carries (Theorem 1); we have not tried to reduce them.

A Trotter simulation. Take an open chain of $n = 64$ spins,

$$H = \sum_{i=1}^{n-1} J_i\,(X_iX_{i+1} + Y_iY_{i+1} + Z_iZ_{i+1}) + \sum_{i=1}^{n} h_iZ_i, \quad (12)$$

with h_i uniform in $[-1,1]$, and either random bonds J_i uniform in $[0.5,1.5]$ or uniform bonds $J_i = 1$. The XX terms commute with each other, as do the YY terms and the ZZ and field terms, so a second-order step is four layers, $XX(\delta t)\,YY(\delta t/2)\,[ZZ+Z](\delta t)\,YY(\delta t/2)$, with the half steps of XX merged across steps. We run to $t = 10$, so $r = 10/\delta t$ steps, at a total synthesis error $\epsilon_{\rm tot}$ of 10^{-3} or 10^{-1}, split evenly over the R approximated rotations of each step: $R = 4(n-1)+n = 316$ with random bonds. With uniform bonds a method may first sum each class of 63 equal angles into its Hamming weight, and then needs one rotation per bit of it, $R = 4\cdot6+n = 88$, at the cost of the Hamming-weight tree; each method, synthesis and gradient alike, takes whichever of the two is cheaper at each step. The Trotter error itself, which sets δt, is outside the comparison; Fig. 4 gives the cost of whatever step is chosen, with batches of at most 256 digits and the gradient's costs averaged over five draws of the roundings. The steps run from 0.3 to 10^{-6}, so the per-rotation errors run from 4×10^{-5} to 3×10^{-13} and the angles from about 0.5 to 10^{-6}: the small-angle regime $\theta^2 \lesssim \epsilon$ is reached only at the smallest steps of the looser budget.

At $\epsilon_{\rm tot} = 10^{-3}$ with random bonds, the gradient layers (sparse pairs) use 0.76 times the T gates of Bothe *et al.*'s

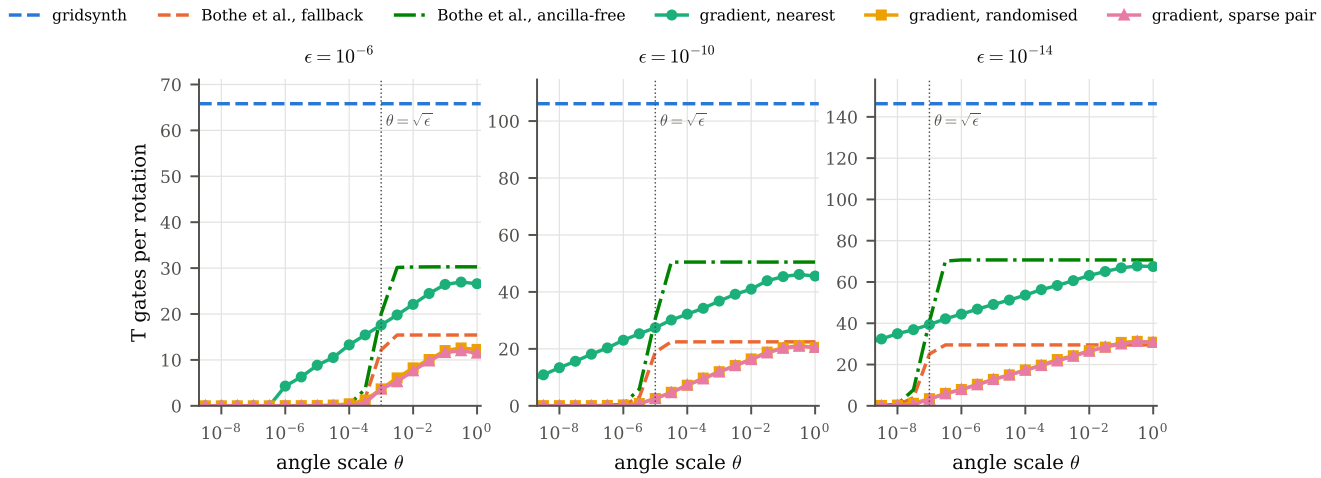


FIG. 2. T gates per rotation in a layer of $L = 1024$ rotations, in batches of at most 256 signed digits, against the angle scale θ (angles log-uniform in $[\theta/2, \theta]$, random signs), at three per-rotation errors. Dashed: gridsynth (fit to runs), flat in θ, and the angle-dependent costs of Bothe *et al.* [9] on the same angles, flat down to $\theta \approx \sqrt{\epsilon}$ and falling steeply below. The gradient's cost falls by about one T gate per halving of the angle over the whole range above $\sqrt{\epsilon}$.

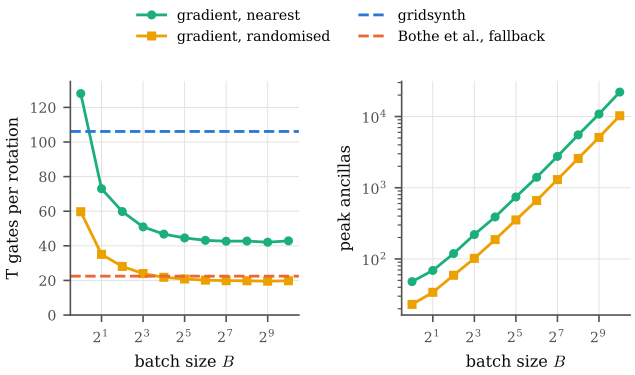


FIG. 3. The batch size. Left: T gates per rotation at $\epsilon = 10^{-10}$, angles in $[0.05, 0.1]$. A batch of one is a single rotation (Proposition 2) and is worse than synthesis; two rotations already beat gridsynth, and sixteen match Bothe *et al.*'s fallback scheme, which at these angles is the mixed fallback cost. Right: peak ancillas, besides the gradient register, which grow linearly with B.

fallback scheme at $\delta t = 0.3$, 0.61 at 3×10^{-3}, 0.34 at 3×10^{-5} and 0.16 at 10^{-6}. At $\delta t = 10^{-3}$ the whole run costs 6.81×10^7 T gates by their fallback scheme, 1.51×10^8 by their ancilla-free scheme, 3.19×10^8 by gridsynth, and 3.41×10^7 with the gradient. With uniform bonds, where Hamming-weight phasing already takes much of the cost, the ratio runs from 0.83 to 0.53. At $\epsilon_{\text{tot}} = 10^{-1}$ the ratios fall to 0.20 with random bonds and 0.39 with uniform bonds; here the smallest steps reach the regime where synthesis falls as θ^2/ϵ, and the ratio climbs back to 0.74 and 0.77 at $\delta t = 10^{-6}$. Over all four cases and all 23 steps the gradient never uses more T gates than Bothe *et al.*'s fallback scheme; the ratio lies between 0.16 and 0.88. Randomised rounding onto adjacent grid points does lose at the smallest steps of the looser budget, by up to a factor of 1.19: there the sparse pairs matter. Sending each rotation to whichever of the two methods has the lower expected cost, judged per rotation before the draw, changed the totals by -7 to $+12$ per cent: the per-rotation estimate misses the shared addition and the Hamming-weight tree. We do not show it; a better router would judge whole batches. Nearest rounding uses 0.24 to 0.51 times the T gates of gridsynth. The ancillas peak at 589 for the randomised layers and 656 for nearest rounding.

VI. DISCUSSION

So the answer to the question we started with is that a phase gradient does not make a rotation cheaper, but it makes a layer of rotations cheaper, because it turns the precision of each angle into a cost per nonzero digit and pays for the carry chain once per layer. The cost of a rotation is then about four T gates per signed digit, so it depends on $\log_2(\theta/\Delta)$ rather than on $\log_2(1/\epsilon)$, and randomised rounding halves the number of digits by halving $\log_2(1/\Delta)$. Angle-dependent synthesis [9] and the gradient exploit small angles in different ranges: synthesis below $\sqrt{\epsilon}$, where a rotation is mostly the identity, and the gradient everywhere above it, where synthesis still pays the full $0.53 \log_2(1/\epsilon)$. Most rotations of a Trotter step at a useful precision are in the second range, and there the gradient wins by a factor of up to six in our examples.

Several things are not in the comparison.

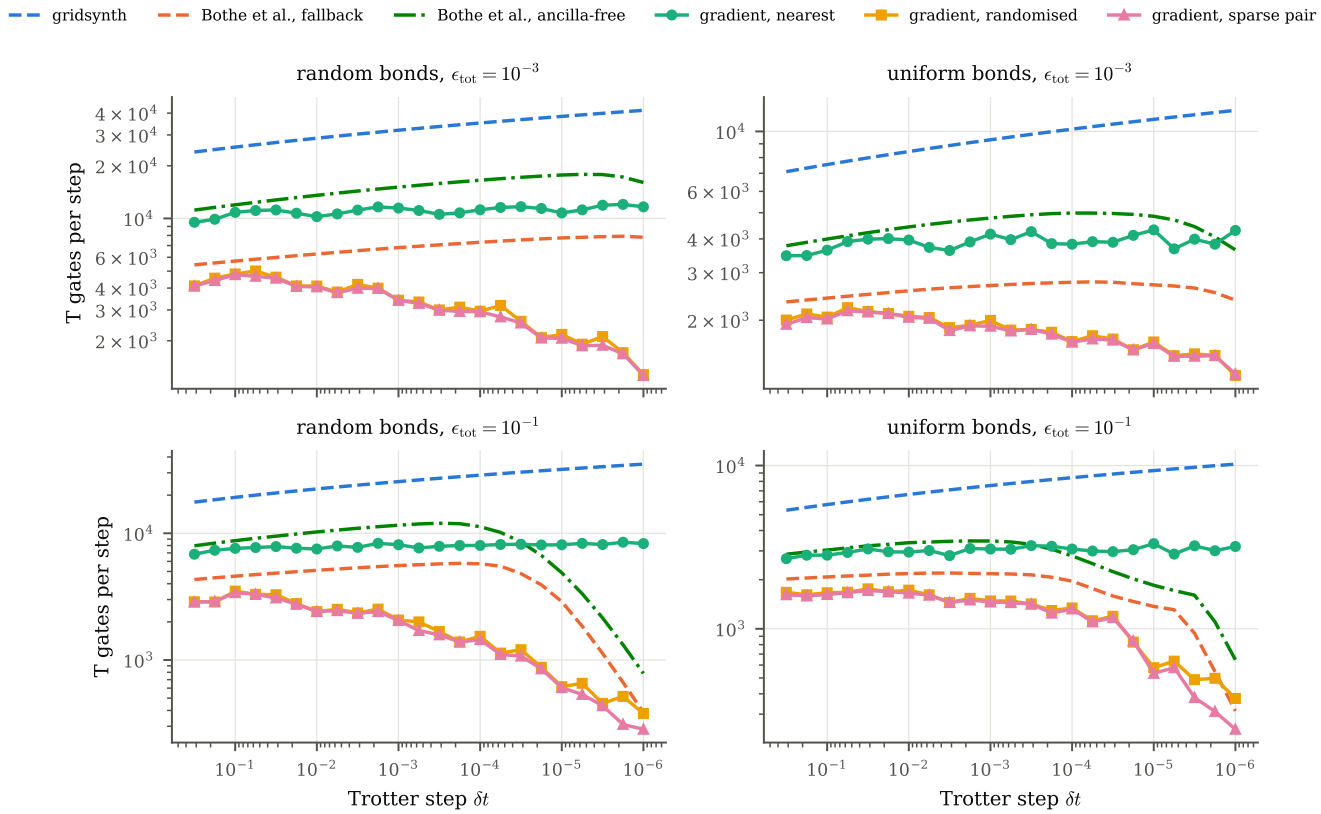


FIG. 4. T gates per second-order Trotter step of a Heisenberg chain of 64 spins in a random field, run to $t = 10$, against the step δt, at total synthesis errors of 10^{-3} (top) and 10^{-1} (bottom). Left: random bonds, every angle distinct. Right: uniform bonds, where each method uses Hamming-weight phasing if it is cheaper. The total is the step cost times $10/\delta t$ steps. Synthesis by Bothe *et al.*'s formulas gets dearer as the step shrinks until the angles reach $\sqrt{\epsilon}$, then cheaper; the gradient layers get cheaper throughout.

- *T gates only.* We count T gates and ancillas. We have not laid out the circuits, scheduled them or counted their depth. The ANDs of a column reduction are sequential within a column; the additions are a carry chain of $b - 2$ Toffolis. Synthesised rotations are sequences of T gates on one qubit, and they parallelise across the layer.

- *Toffolis as four T gates.* With a catalysed CCZ factory a Toffoli costs about as much as two T gates [17], which would roughly halve the gradient's counts. We have not used this.

- *Synthesis costs are formulas.* We evaluate the formulas of Refs. [5, 9] on our angles; we have not run their methods. The fallback scheme's small-angle term is their substitution $\epsilon \to 2\epsilon$ in the ancilla-free one.

- *Quasi-probabilities.* Bothe *et al.*'s largest savings come from quasi-probability mixtures with a total λ of order one, which cost a constant factor in samples instead of an error. Randomised rounding onto the gradient could be given negative weights in the same way; we have not done it, and do not compare in that setting.

- *Ancillas.* A batch of 256 digits needs 500 to 660 ancillas, most of them copies of digit bits and carries kept until the uncompute. Smaller batches trade them for the share of the addition (Fig. 3). A layer whose bits were reduced in place, or whose columns were streamed, would need fewer; we have not tried. Synthesis needs none (or one, for fallback).

- *Equal angles.* When a layer has many equal angles, Hamming-weight phasing does most of the work and the gradient adds less (the uniform chain). Angles that share digits but are not equal, as in a Fourier series of coefficients, could share partial sums across columns; we have not looked at this.

- *The gradient register.* It needs b qubits and $b - 3$ synthesised rotations, once. Its error enters once. We have not included either.

The construction is the oldest trick in multiplier design, a column reduction of partial products [21], put in front of the phase-gradient adder; and its parts, the

measured AND [11, 16], Hamming-weight phasing [11, 14], weighted sums of bits into a gradient [13, 23], and mixing [7, 8], are all known. What we have added is the observation that for fixed angles the reduction can run over signed digits, the count that follows from it, and the comparison with angle-dependent synthesis that shows where it pays: for angles between about $\sqrt{\epsilon}$ and one, once a batch holds a few rotations, which is where most of a Trotter step lies.

Code availability. The code that compiles the layers, simulates them, counts them and makes the figures, with its tests and data, is in Ref. [15], directory `phasegrad_trotter`.

[1] S. Lloyd, Universal quantum simulators, Science **273**, 1073 (1996).

[2] M. Suzuki, General theory of fractal path integrals with applications to many-body theories and statistical physics, J. Math. Phys. **32**, 400 (1991).

[3] A. M. Childs, Y. Su, M. C. Tran, N. Wiebe, and S. Zhu, Theory of Trotter error with commutator scaling, Phys. Rev. X **11**, 011020 (2021).

[4] N. J. Ross and P. Selinger, Optimal ancilla-free Clifford+T approximation of z-rotations, Quantum Inf. Comput. **16**, 901 (2016).

[5] V. Kliuchnikov, K. Lauter, R. Minko, A. Paetznick, and C. Petit, Shorter quantum circuits via single-qubit gate approximation, Quantum **7**, 1208 (2023).

[6] A. Bocharov, M. Roetteler, and K. M. Svore, Efficient synthesis of universal repeat-until-success quantum circuits, Phys. Rev. Lett. **114**, 080502 (2015).

[7] M. B. Hastings, Turning gate synthesis errors into incoherent errors, Quantum Inf. Comput. **17**, 488 (2017).

[8] E. Campbell, Shorter gate sequences for quantum computing by mixing unitaries, Phys. Rev. A **95**, 042306 (2017).

[9] M. Bothe, C. Sünderhauf, M. J. Witham, E. T. Campbell, and N. S. Blunt, More efficient Clifford+T synthesis for small-angle rotations and application to Trotterization, arXiv:2605.31544 (2026).

[10] A. Y. Kitaev, A. H. Shen, and M. N. Vyalyi, *Classical and Quantum Computation* (American Mathematical Society, 2002).

[11] C. Gidney, Halving the cost of quantum addition, Quantum **2**, 74 (2018).

[12] Y. Nam, Y. Su, and D. Maslov, Approximate quantum Fourier transform with $O(n \log(n))$ T gates, npj Quantum Inf. **6**, 26 (2020).

[13] Y. R. Sanders, D. W. Berry, P. C. S. Costa, L. W. Tessler, N. Wiebe, C. Gidney, H. Neven, and R. Babbush, Compilation of fault-tolerant quantum heuristics for combinatorial optimization, PRX Quantum **1**, 020312 (2020).

[14] I. D. Kivlichan, C. Gidney, D. W. Berry, N. Wiebe, J. McClean, W. Sun, Z. Jiang, N. Rubin, A. Fowler, A. Aspuru-Guzik, H. Neven, and R. Babbush, Improved fault-tolerant quantum simulation of condensed-phase correlated electrons via Trotterization, Quantum **4**, 296 (2020).

[15] S. J. Elman, Code and data for "Trotter layers by addition into a phase gradient", https://github.com/samuelelman-blip/Directestimation-of-fixed-time-Fourier-responses (2026).

[16] C. Jones, Low-overhead constructions for the fault-tolerant Toffoli gate, Phys. Rev. A **87**, 022328 (2013).

[17] C. Gidney and A. G. Fowler, Efficient magic state factories with a catalyzed $|ccz\rangle \to 2|t\rangle$ transformation, Quantum **3**, 135 (2019).

[18] D. Aharonov, A. Kitaev, and N. Nisan, Quantum circuits with mixed states, in *Proceedings of the 30th Annual ACM Symposium on Theory of Computing* (1998) p. 20.

[19] G. W. Reitwiesner, Binary arithmetic, Adv. Comput. **1**, 231 (1960).

[20] S. Arno and F. S. Wheeler, Signed digit representations of minimal Hamming weight, IEEE Trans. Comput. **42**, 1007 (1993).

[21] C. S. Wallace, A suggestion for a fast multiplier, IEEE Trans. Electron. Comput. **EC-13**, 14 (1964).

[22] pygridsynth, Python package, version 2.0.0, https://pypi.org/project/pygridsynth/ (2026).

[23] A. Dutta and N. Sarkar, Sum-of-squares compilation of matrix-model dynamics for NISQ and fault-tolerant hardware, arXiv:2610.04731 (2026).